



VNR Vignana Jyothi Institute Of Engineering and Technology

**Department of
Computer Science and Engineering**

TempleYatra

A Gen AI-Powered On-the-Go Temple Guide

Team Members:

- **22071A0502 - Akshith Pesaramalli**
- **22071A0508 - Naga Sai Kiran Battina**
- **22071A0545 - Bhargavi Mukku**
- **22071A0557 - Viswa Prateek Tummala**

Guide

Dr. A. Madhavi

Assistant Professor

Department of CSE & CSBS



Department of Computer Science and Engineering **AGENDA**

1. Introduction
2. Existing System & Limitations
3. Proposed System
4. Problem Statement
5. Objectives
6. Methodology
7. Functional Requirements
8. Work Plan & Distribution of Work
9. References



Department of Computer Science and Engineering

SUGGESTIONS GIVEN BY THE EXTERNAL EXPERT

- **Scalability** – Ensure the system can scale from a few temples in one state (e.g., Tamil Nadu) to pan-India coverage without major redesign.
- **Data Validation** – Create a robust mechanism for verifying community-contributed content to maintain authenticity and prevent misinformation.
- **Performance Optimization** – Use lightweight models or edge inference for TTS/translation to reduce latency on mobile devices.
- **Security & Privacy** – Implement proper authentication and protect user data (location, preferences, travel history).
- **Multilingual Depth** – Go beyond translation by including regional dialects and cultural nuance in storytelling.

Pilot Testing & Validation:

- **Pilot Run:** Test the app with a small group of yatris (students, families, or local devotees) on a short temple route (e.g., 2–3 temples).
- **Quantitative Metrics:** Measure itinerary accuracy, navigation reliability, audio guide playback success rate, and app response time.
- **Qualitative Feedback:** Collect user impressions on storytelling quality, cultural depth, language clarity, and overall ease of use.
- **Identify Pain Points:** Note where users face issues (e.g., wrong timings, poor offline performance, unclear instructions, heavy battery/data usage).
- **Iterative Improvements:** Fix identified gaps in itinerary generation, offline packs, and language translations before wider release.



Department of Computer Science and Engineering INTRODUCTION

Religious tourism in India is one of the largest and most culturally significant forms of travel, yet it remains underserved by intelligent travel technology. Existing pilgrimage options are either rigid, pre-packaged tours with limited flexibility or self-planned trips that often miss out on authentic insights, local storytelling, and contextual guidance. For modern spiritual seekers, there is a growing need for a solution that combines autonomy in travel with deeper cultural engagement and real-time support.

TempleYatra addresses this gap by providing a Gen AI-powered, multilingual, on-the-go temple guide for personalized theertha yatras. The platform leverages advanced AI to generate optimized itineraries, deliver rich cultural storytelling, and offer real-time route adjustments based on travel conditions. With features such as offline-ready temple packs, regionally accented audio guides, and a community-driven contribution engine, TempleYatra empowers spiritual travelers to explore sacred destinations with authenticity, flexibility, and ease while preserving and enriching local traditions.



Department of Computer Science and Engineering EXISTING SYSTEM & LIMITATIONS

- Existing devotional apps center on darshan, e-puja, and content; they don't provide end-to-end, on-trip planning and live guidance for multi-temple yatras.
- Audio guide/tour apps are static and site-specific; they lack dynamic rerouting, temple timing logic, or festival/crowd-aware adjustments during travel.
- Tour marketplaces optimize pre-trip bookings but not in-journey re-optimization across transport, weather, closures, and delays.
- Services are fragmented across devotion, routing, booking, and lodging—forcing app hopping instead of a unified pilgrim assistant.
- Offline reliability is inconsistent for remote corridors; downloadable, multilingual temple packs are limited or shallow.
- Cultural depth and validation are weak; sthala puranam and local lore aren't systematically digitized, geo-linked, multilingual, and admin-verified.



Department of Computer Science and Engineering PROPOSED SYSTEM

- **Dynamic, constraint-aware yatra routing that auto-reorders multi-temple itineraries using live signals from transport, temple timings, weather, and festival congestion for minimal disruption and budget adherence.**
- **Multilingual RAG storytelling that fuses structured temple data with digitized folklore; delivers culturally accurate, regionally accented audio guides and translations on-device/edge where feasible.**
- **Unified pilgrim assistant integrating maps, transit, accommodations, festival calendars, and weather into a single workflow with proactive alerts and decision support during the journey.**
- **Robust offline “temple packs” bundling maps, audio guides, rituals, dress codes, and emergency info; optionally powered by local/offline RAG and TTS for low-connectivity corridors.**
- **Community contribution and admin validation pipeline to capture priests’ stories and pamphlets, apply OCR and curation, geo-link to temples, and maintain provenance and sensitivity compliance.**



VNR Vignana Jyothi Institute Of Engineering and Technology

**Department of
Computer Science and Engineering**

PROBLEM STATEMENT

A unified, accessibility-aware, multilingual, offline-ready assistant that dynamically plans and re-optimizes multi-temple yatras while delivering culturally authentic storytelling—solving today’s fragmented, static, and crowd-insensitive pilgrimage tools

A Gen AI-Powered On-the-Go Temple Guide



Department of Computer Science and Engineering

OBJECTIVES

1. **Deliver dynamic, constraint-aware yatra planning that re-optimizes multi-temple itineraries in real time using transport, temple timings, weather, and festival signals.**
2. **Provide culturally authentic, multilingual storytelling (sthala puranam, rituals, mantras) via RAG and high-quality, regionally accented audio.**
3. **Unite maps, transit, accommodations, crowd/festival calendars, and safety into one assistant with proactive alerts and offline-ready “temple packs.”**
4. **Build a community contribution and admin validation pipeline to digitize, geo-link, and govern local lore with provenance and accessibility-first UX.**



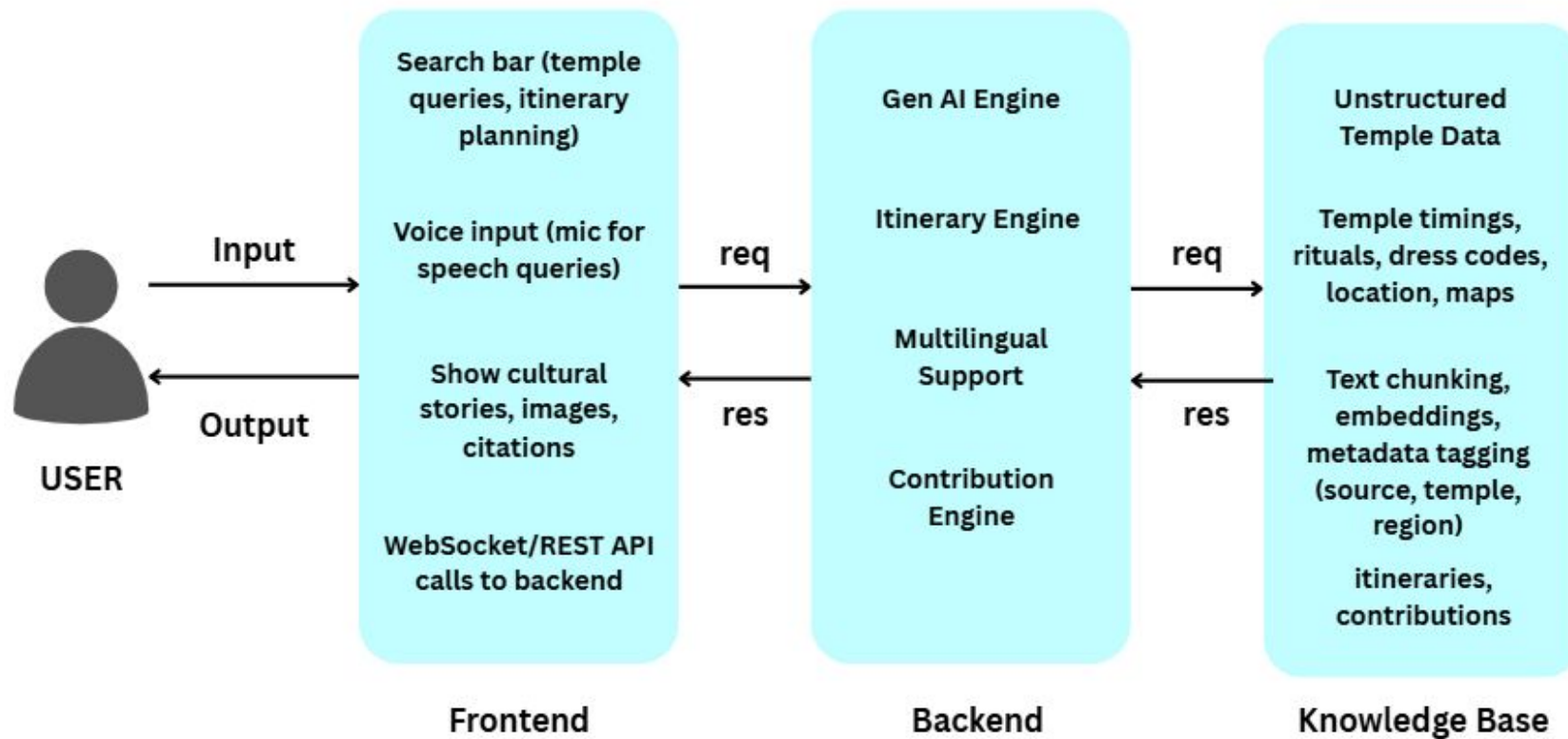
Department of Computer Science and Engineering

METHODOLOGY

- **Data acquisition and curation:** Collect temple data (timings, rituals, histories), festival calendars, transport schedules, weather, and local pamphlets; apply OCR, cleaning, metadata tagging, and geo-linking to temples for a structured knowledge base.
- **Multilingual content pipeline:** Build RAG-ready corpora; fine-tune translation and TTS for Indian languages with regional accents; validate cultural accuracy with domain reviewers and provenance tracking.
- **Dynamic routing engine:** Implement constraint-aware itinerary planning that ingests live signals (transport delays, temple closures, crowd/festival intensity, weather) and re-optimizes routes with user budget/time constraints.
- **Unified integrations:** Connect maps/navigation, transit, accommodations, and alerts into a single flow; standardize via API gateways and retries, with fallbacks for partial data or outages.
- **Offline-first delivery:** Package “temple packs” (maps, audio, rituals, dress codes, safety info) with on-device search, lightweight embeddings, and TTS; sync deltas when online for minimal bandwidth use.
- **Community and governance:** Enable story capture (audio/text), temple mapping, and annotations; run moderation with admin review, source provenance, and sensitivity compliance; measure quality with feedback loops and A/B tests.



Department of Computer Science and Engineering ARCHITECTURE





Department of Computer Science and Engineering MODULES

- **Itinerary & routing:**
Dynamic, constraint-aware planner that auto-reorders multi-temple routes using live temple timings, transport, weather, and festival congestion.
- **Cultural storytelling:**
Multilingual RAG module delivering temple-specific sthala puranam, rituals, and mantras with regionally accented audio and citations.
- **Integrations & offline packs:**
Unified maps/transit/hotels/festival APIs with downloadable “temple packs” for maps, audio, dress codes, and safety info in low-connectivity areas.
- **Community & governance:**
Contribution, moderation, and provenance module to ingest priest stories and pamphlets, apply OCR/geo-linking, and enforce sensitivity and accessibility standards.



Department of Computer Science and Engineering FUNCTIONAL REQUIREMENTS

1. Itinerary & routing:

Use India-focused routing/traffic APIs (Mappls/MapmyIndia, Ola Maps) plus a constraints engine that ingests temple timings, service windows, and travel-time uncertainty; run periodic re-optimization with ETA deltas and festival crowd signals to reshuffle stops within budget/time bounds.

2. Cultural narration:

Build a multilingual RAG pipeline using cross-lingual embeddings, language detection, and translation layers.

3. Integrations & alerts:

Orchestrate maps, transit, accommodations, weather, and festival calendars behind an API gateway with retries and circuit breakers; push actionable notifications when conflicts arise, offering one-tap alternatives and re-routing.

4. Offline temple packs:

Package local vector stores and audio assets; enable on-device retrieval with lightweight DBs (Faiss/Qdrant Lite) and quantized LLMs for offline Q&A; sync diffs when online to update timings and safety info efficiently.

5. Community & moderation:

Implement an in-app submission flow with OCR and geo-linking; add automated toxicity/PII checks, human review queues, and audit logs; publish clear community guidelines and a reporting system with escalation SLAs.



Department of Computer Science and Engineering PLAN OF WORK & WORK DISTRIBUTION

Phase 1: Data Collection & Processing (Weeks 1–5)

- Gather temple data (websites, PDFs, pamphlets).
- OCR for scanned/local notes.
- Preprocess, clean, and structure into database.

Phase 2: Knowledge Base & Retrieval (Weeks 6–10)

- Text chunking + embeddings.
- Vector DB setup (FAISS/Pinecone).
- Build retrieval pipeline with basic multilingual support.
-

Phase 3: Response & Assistance (Weeks 11–15)

- Connect retrieval with LLM for Q&A and itinerary generation.
- Add cultural storytelling + multi-turn handling.
- Integrate APIs (Maps, Weather, Festivals) for dynamic adjustments.

Phase 4: Advanced Features & Finalization (Weeks 16–20)

- Develop offline-ready temple packs.
- Implement audio guides (TTS, regional accents).
- Enable community contributions + admin validation.
- Testing, bug fixes, and documentation.



Department of Computer Science and Engineering REFERENCES

- [1] M. M. Dagli, S. Ahmed, J. Kim, and K. Lee, "Development and validation of a novel AI framework using rule-based NLP and LLM integration for surgical data extraction," *Scientific Reports*, vol. 14, no. 11, pp. 77535, Nov. 2024. doi: 10.1038/s41598-024-77535-y. [Online]. Available: <https://www.nature.com/articles/s41598-024-77535-y>
- [2] A. Chen, X. Ge, Z. Fu, Y. Xiao, and J. Chen, "TravelAgent: An AI Assistant for Personalized Travel Planning," *arXiv preprint*, arXiv:2409.08069, 2024. [Online]. Available: <https://arxiv.org/abs/2409.08069>
- [3] J.-H. Wu, S.-D. You, and M.-J. Lo, "Interactive Virtual Reality Touring System: A Case Study of Shulin Ji'an Temple in Taiwan," *Advances in Human-Computer Interaction*, vol. 2021, Art. 6651916, 2021. doi: 10.1155/2021/6651916. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1155/2021/6651916>
- [4] Y. Zang, "TripTrail: AI-Based Smart Travel Planner," *Journal of Emerging Technologies and Innovative Research*, vol. 12, no. 5, May 2025. ISSN: 2349-5162. [Online]. Available: <https://www.jetir.org/papers/JETIR2505743.pdf>
- [5] Z. Zhang, "Enhancing Intelligent Voice Assistants in Tourism Services: A Computational Approach," in *Proc. 2024 Int. Conf. on Artificial Intelligence and Management*, pp. 1–8, 2024. [Online]. Available: <https://www.atlantispress.com/article/126007174.pdf>
- [6] A. P. Wibawa, "Advancements in natural language processing," *Journal of Artificial Intelligence and Data Science*, vol. 2, no. 1, pp. 59–72, 2024. doi: 10.1016/j.jaid.2024.05.009. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2772503024000598>
- [7] E. V. Azevedo, "Explained: What is Retrieval-Augmented Generation (RAG)," *HiJiffy Technical Whitepaper*, Jun. 2025. [Online]. Available: <https://www.hijiffy.com/resources/articles/explained-retrieval-augmented-generation-rag>



Department of Computer Science and Engineering

[8] S. Schmidt, et al., “Data extraction methods for systematic review (semi-automated),” *PLOS Medicine*, Apr. 2025. PMC8361807. [Online]. Available:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8361807/>

[9] L. D. Wang, “The Value and Application of Multilingualism in the High-Quality Development of Smart Tourism,” *Journal of Service Science and Management*, vol. 18, no. 5, pp. 378–389, May 2025. doi: 10.4236/jssm.2025.185027. [Online]. Available: <https://www.scirp.org/journal/paperinformation?paperid=143085>

[10] S. S. Batchelor, S. Dastidar, and P. M. Mishra, “Empowering Multilingual Forms with Generative AI using MarianMT and T5 Models,” *Computer Science & Information Technology (CSIT)*, vol. 14, pp. 65–75, 2025. [Online]. Available: <https://aircconline.com/csit/papers/vol14/csit141305.pdf>

The slide features several decorative elements: a purple circle in the top-left corner, a green L-shaped line in the top-right corner, an orange circle in the middle-left area, a green L-shaped line in the bottom-left corner, and an orange circle with three blue curved lines above it in the bottom-center.

THANK YOU